

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED Intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

Binay Opto Electronics Pvt. Ltd.

27, PANDITIA TERRACE, Calcutta 700 029, India
Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030
email: info@binayled.com
Fax: 91-33-2242 1493 <http://www.binayled.com>

POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

interference, such WLAN networks can also reach other communication partners across several kilometres.

According to researchers, networks will be available for all passers-by on nearby streets so that data can be transferred to and from their smartphones.

Scientists light the way for future electronic devices

Researchers from Optoelectronics Research Centre (ORC) at University of Southampton have demonstrated how glass can be manipulated to create electronic devices that are smaller, faster and consume less power.

Working in collaboration with Advanced Technology Institute at University of Surrey and University of



Dr Behrad Gholipour with his prototype device

Cambridge, ORC scientists Dr Jin Yao and Dr Behrad Gholipour have made prototype devices that use light to bring together different computing functions into one component.

Silicon is reaching its fundamental limits and soon we

will be unable to make the silicon chip any smaller or decrease the power consumption of these devices. Using a family of materials called chalcogenides, the team is looking for the next generation of materials, which could replace traditional semiconductors and conventional electronics that power today's electronic devices.

By doping a chalcogenide glass, in this case a material based on germanium and selenium, and forming a multi-layer structure with electrical contacts, Dr Gholipour was able to show the same switching behaviour as seen in a silicon transistor, but in a device made entirely from glass.

He explained, "Non-equilibrium doping within chalcogenide glasses enables a unique information-processing platform within one material system. This allows traditional electronic computing, along with memory functionality, which opens up the optical spectrum, from the visible far into the infra-red, for next-generation optoelectronic and fully-optical computing applications."

This work builds on Dr Behrad's PhD work for which he won UK Engineering and Physical Sciences Research Councils (EPSRC) prestigious ICT pioneer prize in the area of information in 2011. Using similar materials, Dr Yao formed long strips of different types of chalcogenides, connecting each end together to form a chain of materials.

Dr Yao said, "When one end of the structure is hotter than the other, as might occur near the battery of your phone or computer, a voltage is produced through a process known as the thermoelectric effect. This has the potential to make our electronic devices more efficient."